iregi-rust benchmark

Batched one-loop tensor reduction in Rust versus IREGI and MadLoop

2026-10-23 · AMD EPYC 9754 · CPU 0 · historical 4c32ece baseline

Reading guide. Every timing ratio is *iregi-rust time ÷ reference time*. Green means faster, white parity and red slower; timings are per phase-space point. Gray cells are unmeasured and amber cells are pending verification. Source inspection does not establish numerical agreement. Both historical 4c32ece backends pass all 8 numerical checks. Interpreter: 7 regressions and 1 improvement. Native: 5 regressions, 1 parity result and 2 improvements. Each uses its own paired IREGI run; the performance goal remains unmet and newer changes are unmeasured.

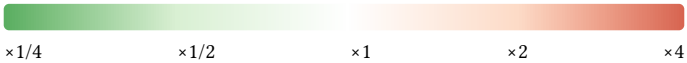
Acceptance gates

Correctness	Agreement with original IREGI and isolated MadLoop graphs	**Pending**
Kernel performance	Equal or better on identical inputs; historical baseline	**Unmet in both runs**
Complete evaluation	Report preparation, numerator, reduction and masters separately	**Unmeasured**

1 Historical kernel baseline: 4c32ece

Table 1: **Massive bubble/triangle, binary64, hybrid mode.** Rust and IREGI total medians in µs/point. Ratios are paired-block medians; intervals are 99% bootstrap confidence intervals. Batch 1 is latency; batch 32 is amortized cost per point. Each backend uses its own paired reference.

Case	Rank	Interpreter Batch 1	Interpreter Batch 32	Native Batch 1	Native Batch 32
Bubble	0	**3.176 µs** IREGI 1.988 µs · ×1.598 99% CI 1.585–1.607	**1.708 µs** IREGI 1.885 µs · ×0.906 99% CI 0.902–0.911	**0.936 µs** IREGI 0.943 µs · ×0.993 99% CI 0.987–1.003	**0.903 µs** IREGI 0.878 µs · ×1.031 99% CI 1.024–1.042
Bubble	1	**9.552 µs** IREGI 5.707 µs · ×1.670 99% CI 1.655–1.688	**5.831 µs** IREGI 5.748 µs · ×1.015 99% CI 1.008–1.019	**4.663 µs** IREGI 3.044 µs · ×1.534 99% CI 1.526–1.549	**3.231 µs** IREGI 3.029 µs · ×1.069 99% CI 1.062–1.081
Triangle	0	**29.152 µs** IREGI 8.496 µs · ×3.446 99% CI 3.415–3.454	**18.229 µs** IREGI 8.433 µs · ×2.160 99% CI 2.147–2.172	**4.227 µs** IREGI 4.968 µs · ×0.849 99% CI 0.844–0.858	**8.405 µs** IREGI 4.899 µs · ×1.715 99% CI 1.704–1.723
Triangle	1	**38.505 µs** IREGI 18.297 µs · ×2.101 99% CI 2.088–2.108	**25.634 µs** IREGI 18.224 µs · ×1.408 99% CI 1.400–1.413	**8.809 µs** IREGI 10.418 µs · ×0.850 99% CI 0.844–0.854	**12.461 µs** IREGI 10.213 µs · ×1.220 99% CI 1.206–1.228



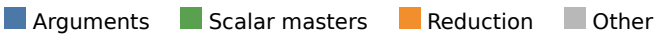
Both backends pass all eight numerical checks at 10^{-10}. The interpreter has seven regressions and one improvement; native has five regressions, one parity result and two improvements. **Absolute IREGI timings changed substantially between sessions**, despite identical inputs/reference binary and CPU settings. Each verdict uses its own paired run; no direct backend speedup is inferred across sessions. Both use the historical `4c32ece` revision; current-tree, higher-rank and physical performance remain unmeasured.

Newer validation includes four integrated graphs. The top pentagon at Float128/256 agrees with PV/hybrid and MadLoop; original IBP still fails. Physical timings are unmeasured.

2 Historical stage profiles: 4c32ece

Rust medians from separate instrumented passes, in µs/point. All bars use 0–40 µs/point; S denotes the Symbolica interpreter, N the generated native backend. Sessions are separate: compare each with its own reference on page 1, not with the other backend's absolute bar length. Raw ns/batch are divided by 1000 × batch width.

Case / batch / backend	Rust stage time: 0 20 40 µs/point	Total
Bubble r0 / 1 · S		3.311
Bubble r0 / 1 · N		1.057
Bubble r0 / 32 · S		1.708
Bubble r0 / 32 · N		0.900
Bubble r1 / 1 · S		9.754
Bubble r1 / 1 · N		4.799
Bubble r1 / 32 · S		5.847
Bubble r1 / 32 · N		3.218
Triangle r0 / 1 · S		29.251
Triangle r0 / 1 · N		4.333
Triangle r0 / 32 · S		18.114
Triangle r0 / 32 · N		8.369
Triangle r1 / 1 · S		38.602
Triangle r1 / 1 · N		8.917
Triangle r1 / 32 · S		25.595
Triangle r1 / 32 · N		12.426



"Other" is total median minus named stage medians, not a separately timed region. Scalar-master time includes complete master calls; reduction includes stability checks. IREGI has no measured stage split here. For native triangle rank 0, scalar masters cost 3.282 µs at batch 1, but 249.977 µs per batch 32, or 7.812 µs/point: this batch path has higher amortized master cost. The observation identifies a profiling target, not a cross-session speedup or a proven cause. Full stage values and samples remain in the raw data.

2.1 Preparation and measurement conditions

Backend	Bubble r0	Bubble r1	Triangle r0	Triangle r1
Interpreter	1.572 s	0.143 s	0.104 s	0.091 s
Native	0.011 s	0.012 s	0.001 s	0.013 s

Preparation values are single observations without repeated uncertainty, excluded from warm kernel ratios. Batch32 reuses the plan; numerator preparation is outside this tensor-array benchmark.

Both backends of the isolated pushed revision `4c32ece` ran on AMD EPYC 9754 CPU 0, with 5 warm-ups and 21 paired blocks targeting 30 ms, randomized order with seed 271828. Rust used release opt-level3, thin LTO and one codegen unit; Fortran used `-O3`. Structural plans and allocation capacity were reused; numerical results were recomputed, and original recurrence caches were freed per sample. Subprocess transport is excluded. All source/binary provenance checks pass, and every ratio interval is no wider than 2% of its median. Raw samples, inputs, flags and hashes are retained in `validation/benchmark/baseline-4c32ece/` and `validation/benchmark/native-4c32ece/`.

A separate frozen-native policy experiment keeps `Whole`: all 18 C0/D0 comparisons across three orderings find `FixedRows4/8/16` slower, with per-comparison 99% intervals excluding parity. It uses 32 valid lanes, CPU 0 and 21 rotated blocks. This selects a batch policy, not IREGI speedup; see `validation/native-policy/`.

3 Validation coverage

Table 2: **Evidence ledger.** “Source audited” records code inspection; “verified” identifies executed checks described in that row.

Capability	Required evidence	Status
Independent reference oracle	Untouched/repaired Fortran twins 24 ordinary cases identical; 80 table entries exact; 7 generalized fixtures pass. New pentagon audits reveal reference defects.	**Verified**
Scalar-master component	OneLoopMaster A0/B0/C0/D0/DB0 Final27 scalar checks and13 consolidated D6 checks pass; all5 mixed physical D0 cases pass atFloat384. Unsupported derivative domains remain explicit.	**Verified**
IBP component	Powers, dimension shifts and singular identities 21 exact tests; the independent Gamma oracle covers 200 tensor requests and 40 generalized requests in all four slots.	**Verified**
PV component	Every tensor moment and native Laurent slot 10 exact tests; N1–4 ranks0–7 pass. N5 rank6: 840 slots converge at128/256 bits. Rank7 preparation timed out; no coefficients compared.	**Partial**
Precision and Dual	binary64, DoubleFloat, Float, static/dynamic jets f64,DoubleFloat,Float128/256/512,static/dynamic Dual and selected physical mixed jets pass; fresh mixed Float80/256 argument guards pass. Broader recurrence precision audit remains open.	**Partial**
Execution / CLI / core oracle	Batches through reduction and scalar masters 30 execution and5 argument-guard tests pass. CLI19 smoke checks pass; frozen31 fixtures include30 binary64 plus1 explicit verified rescue.	**Partial**
Exceptional kinematics	Explicit errors and verified fallback 15 stability checks and full N5 rank6 rescue pass. Legacy corpus:22 primary plus8 verified128/256-bit rescues pass all30 retained cases.	**Partial**
Graph/numerator preparation	FeynKit DOT to Symbolica coefficients 4 model-boundary, 13 preparation, 4 physical numerator and 19 CLI smoke checks pass.	**Verified**
Physical MadLoop comparison	Graph-isolated top loops and massless box Integrated ggH, ggHH, ggHHH and massless up-box pass their retained MadLoop comparisons. ggHHH Float128/256 also agrees with original PV/hybrid; original IBP still disagrees and all-mode reports remain failed.	**Partial**

Snapshot boundary. Counts describe the executed component, public-execution and selected physical checks below. They do not certify all kinematics, higher-point ranks or every physical graph. Open discrepancies remain listed with the original tolerance. Master precision checks include binary64, DoubleFloat and Float at 128, 256 and 512 bits. Verified jets include static/dynamic second-order tadpoles and Euclidean massive triangle/box scaling derivatives. Four physical triangle cases and all five massive/complex box cases pass six-component mixed derivative checks. Earlier 180-second box timeouts remain historical evidence. **Verified precision rescue:** for the five-propagator rank-six case, 128- and 256-bit evaluation converge. Binary64 Rust and original IBP differ from that value by 2.23×10^{-10} and 3.40×10^{-10} respectively. Both exceed the fixed 10^{-10} criterion. The public rescue now verifies all 210 moments / 840 slots, with an independent coefficient assertion. The frozen oracle passes all 31 cases: 30 at binary64 and one generalized triangle through explicit 128/256-bit rescue. Broader high-point ranks and physical domains remain pending.

3.1 Known original-reference defect

For shifts $(0,0,0,0)$ and $(0,1,0,0)$, squared masses $(4,9)$, $\mu = 3$ and numerator q_x^4, original PV and hybrid report stability but disagree with an independent Feynman-parameter integral. Their UV residues agree.

Method	Finite coefficient / i
Parameter integral	45.4316300015058951832997…
Original IBP	45.43163000150602
Original PV / hybrid	46.11123944517427

Evidence: `validation/pv-high-rank/evidence.json`. The original PV branch is retained unchanged; the relevant Rust comparisons use original IBP and the independent parameter-integral result.

4 Measurement protocol

Run release builds on one pinned CPU core with fixed thread counts and record hardware, operating system, toolchain, compile flags, dependency revisions, input hashes, seeds, warm-up, sample count and raw observations. Every result must identify whether caches are cold or warm. Report median and dispersion over repeated sufficiently long timed blocks; retain an output checksum.

Preparation comprises graph import, rules, tensor identities, coefficient extraction and evaluator construction. Time numerator evaluation, pure reduction, scalar masters and complete evaluation independently. Measure the complete total directly: adding separately instrumented components is only a diagnostic. Distinguish reference-library calls from subprocess and file I/O.

Use the same samples at batch widths 1, 4, 8, 16, 32 and 64 and report time per point. For precision comparisons, match the numerical precision and include rescue costs. Record failed lanes; do not omit unfavorable points silently. Absolute/scaled coefficient errors, high-precision adjudication and derivative checks accompany performance results. No speed gate passes without correctness. The stable binary64 target is a mixed absolute/relative tolerance of 10^{-10} on dimensionally normalized outputs. Gate singleton latency and batch-32 throughput separately for each topology/rank/mass family. Record IBP, PaVe and hybrid mode independently. The eventual reference envelope is seven propagators and rank seven, including generalized powers and dimension shifts.

5 Rebuilding this report

Update `docs/data/results.json` only from retained measurements and rebuild with `typst compile docs/benchmark.typ docs/benchmark.pdf`. The report uses Libertinus Serif and DejaVu Sans/Mono. Supply their installation directories through Typst's `--font-path` option when needed. Reference checkout revisions are retained in the JSON data. They record the audited source trees and must not be mistaken for a verified dependency build.

5.1 Validation reproduction and remaining checks

The standalone checks reported here run with the pinned dependencies prepared by `scripts/env.sh`. The master suite uses `RUST_MIN_STACK=67108864 target/iregi-masters-tests --test-threads=1 --nocapture` the IBP suite uses `target/ibp-unit --test-threads=1 --nocapture`. Thirty execution and five argument-guard checks pass, including mixed Float80/256, underflow, false/nonzero distance cancellation and higher Taylor components. All nineteen CLI smoke checks pass. The frozen oracle passes 31 cases: 30 at binary64 and `C0_power2_1_1_rank2` through explicit batched 128/256-bit rescue with unchanged tolerance. Three active rescue tests and the complete public rank-six pentagon rescue pass. The primary binary64 lane is correctly flagged unstable; 840 coefficients converge at 128/256 bits and the independent coefficient assertion passes. Four exact bubble-Taylor tests and three supplied-product scalar API tests pass. All 30 retained legacy-snapping cases now pass: 22 at primary precision and eight through explicit 128/256-bit rescue. Earlier failures remain recorded; broader/generalized policy cases are unqualified.

The final frozen master source passes 27 checks, recorded with logs in `validation/scalar-masters/qualification.json`. Independent B0 checks cover kinematic scales 2^{-1070} and 2^{1000} at $\mu^2 = 1$ for f64, DoubleFloat and Float256. Six-component scale-only jets pass for all five master families. Mixed input/seed precision, physical B0/C0 derivatives and subnormal normalization pass, including the complete mixed D0 fixture batch. Native kernels remain opt-in, with the earlier all-family and tail parity checks.

An independent Symbolica Gamma expansion checks 200 tensor and 40 generalized requests with analytic bubbles both disabled and enabled: 480 requests and 1920 native slots for the massless one-off-shell triangle. Nine extracted FeynKit partial-fraction tests pass; planner integration is pending. Two planar formula tests pass, but public activation awaits exact rank certification. These checks establish no broader numerical or performance result.

Exact PV coefficients with original Fortran scalar leaves pass the N1–4 corpus: 330 moments through rank seven, all native slots. N6 binary64 with those leaves has 430 discrepancies. Separately, inline native Rust masters at 128/256 bits converge for all 1320 complex slots within 6.46×10^{-29}; original IBP has 566 failures against that result, and retained binary64 values also fail. This is internal consistency, not public workspace qualification. Public N6/N7 and routing variants remain pending; see `validation/pv-high-rank/rust-coverage.json`. N5 rank-six public rescue is qualified separately. An experimental Symbolica patch now verifies all 1320 N5 rank-seven slots; production integration remains pending. Earlier N5 240 s and inline N7 90 s preparation timeouts remain in `data/results.json`.

6 Physical fixture qualification

Single graph	Pointwise numerator	Integrated Rust result
Top triangle: $gg \to H$	**Verified**	**Verified**
Top box: $gg \to HH$	**Verified**	**Verified**
Top pentagon: $gg \to HHH$	**Verified**	**PV/hybrid + MadLoop**
Massless up-quark box	**Verified**	**Verified**

6.1 Physical reference contract

Fixtures preserve parameters, actual HELAS polarizations, graph factors and source hashes. The triangle uses two orientations with grouping disabled. MadLoop shifts are $(0, p_1, p_1 + p_2)$ with polynomial variable $x = -Q$; its adapter includes the odd-rank sign. For the FeynKit routing, $N_{\mathrm{FK}(q)} = -\frac{1}{2} P_{\mathrm{ML},1}(p_1 - q)$ with matched couplings/polarizations. Do not apply color or direction factors twice.

Five schema-2 captures under `validation/physical/madloop/fixtures/` contain 40 graph records, including a second triangle helicity. All thirteen preparation tests and four selected physical numerator comparisons pass; this is not 40 integrated validations. Actual evaluated momenta are exported because MadLoop can correct the requested phase-space point. The compact triangle reference is `validation/physical/ggh-numerator-reference.json`.

The recorded binary64 $gg \to H$ CLI passes all 15 moments / 60 slots against original IREGI within 1.86×10^{-15} normalized error and MadLoop within 9.38×10^{-15}, including poles. Its fixed-color raw finite value is -147.86843349622455, normalized to -0.9363877940734862. Values and executable provenance are retained in `validation/physical/ggh-cli-comparison.json` this is not a fresh replay. All nineteen current CLI smoke checks pass. Reference data and comparison arithmetic are binary64; higher working precision certifies no extra digits. Short unpinned timing checks are excluded from benchmark evidence. The top-box binary64 rejection remains recorded: PV fails its cancellation guard and IBP fallback is nonfinite. Complete Float128 and Float256 CLI reruns now pass all 280 tensor slots against all three reference modes, within 1.15×10^{-13}; MadLoop contraction error is at most 4.77×10^{-15}. Each run freshly prepares/evaluates the numerator. Their normalized tensor difference is 6.09×10^{-37}, with identical contracted values after binary64 conversion. That comparison arithmetic certifies no extra digits. Evidence is retained in `validation/physical/gghh-cli-precision-convergence.json` and its linked Float128/256 reports. A model-import regression was resolved by a narrow Symbolica-backed quantum- number adapter. Four boundary, thirteen preparation and four physical numerator tests pass with the fix; other numerical tokens retain their precision.

The current optimized CLI also passes the massless up-quark box. All 70 moments / 280 slots agree with each original mode: maximum error 2.85×10^{-14} against IBP and 7.11×10^{-15} against PV/hybrid. The isolated MadLoop finite/single/double coefficients agree within 6.74×10^{-15}, with fixed-color factor $-\frac{1}{8}$. Evidence is retained in `validation/physical/gggg-up-cli-comparison.json` its one-run wall time is not a qualified performance measurement.

Complete top-pentagon CLI reruns at Float128/256 now agree with original PV/hybrid for all 280 slots within 4.63×10^{-14}, and with MadLoop within 9.30×10^{-14}. Between-run tensor difference is 7.79×10^{-17} under binary64 comparison arithmetic. Original IBP still differs by 0.60014, so the all-mode report remains failed. See `validation/physical/gghhh-cli-precision-convergence.json` and linked reports. Earlier PV/hybrid replay of 378 captured MadLoop slots was exact.

6.2 Physical derivative qualification

Four triangle cases at 384 bits cover real subthreshold/absorptive and complex-mass kinematics, each with six first, second and mixed Dual components. Finite differences converge with maximum final scaled error 5.47×10^{-28}. One equal-complex D0 case also passes, with error 1.11×10^{-34}. The complete five-case D0 batch now passes at 384 bits, maximum final scaled error 2.53×10^{-30} under the unchanged 2^{-70} criterion. Its 224.83 s validation duration is not a benchmark; earlier 180 s timeouts remain retained. See `validation/scalar-masters/physical_box_full.json`. Exact commands and scope are retained in `crates/iregi-masters/tests/validation_evidence.json`.

7 Open qualification ledger

Unresolved item	Observed evidence and remaining check
Binary64 baseline accuracy	Historical Rust error 2.23×10^{-10} and original IBP error 3.40×10^{-10} fail the unchanged gate. Public128/256-bit rescue now passes at the retained rank-six point.
Broader pentagon rescue	Rank six passes 840 slots. An isolated Symbolica patch verifies all1320 rank-seven slots at128/256 bits and one independent coefficient. Production integration and broader domains remain pending.
Physical pentagon reference	Complete Float128/256 graph evaluations pass PV/hybrid and MadLoop. Original IBP still disagrees; all-mode reports remain failed. Binary64 and broader kinematics remain unqualified.
Shifted scalar domain	Thirteen consolidated master/endpoint checks and retained public finite fixtures pass. Unsupported endpoints are rejected; public DD/Float batch-3 timed out at 150 s. Broader domains/native E6 remain open.
Broader derivative domains	The full five-case mixed D0 batch now passes with error below 2^{-70}. True thresholds and unsupported transverse massless/Gram/Cayley domains retain typed errors; old timeouts remain.
Physical top-box binary64	The original binary64 guard rejection remains visible. Complete Float128/256 evaluations now pass; this does not establish binary64 stability or performance.
Supplied-product high points	Pure PV/hybrid scalar requests with six or more supplied-product propagators require domain validation. The public certificate wrapper timed out at 90 s. Bounded argument guards pass; broader mixed-precision recurrence conditioning remains unqualified.
Legacy snapping scope	All30 retained cases pass:22 primary plus eight verified128/256-bit rescues. Earlier failures are preserved; generalized requests and broader domains remain unqualified.
Runtime performance	Interpreter: 7 regressions/1 improvement. Native: 5 regressions/1 parity/2 improvements, in a separate paired session. Newer and broader performance remains unmeasured; the goal is unmet.

7.1 Canonical physical reference audit

Original IBP and PV/hybrid disagree in both UV and finite rank-four slots for $gg \to HHH$, despite stable flags. The largest normalized finite error is 0.6001338886545732 at moment 54. UV power counting and converged high-precision evaluation now adjudicate that component; details follow on the next page. The other three reference families agree across original modes within 2.22×10^{-14}. All-slot comparisons are retained in `validation/physical/canonical-reference-audit.json`.

The retained `validation/pv-high-rank/reference-coverage.json` also records rank-seven original PV instability for five to seven propagators and a falsely stable four-propagator disagreement. These reference failures remain visible; stable IBP reference values do not themselves certify Rust high-rank coverage.

7.2 Independent integral evidence

The rank-four bubble's 50/80-digit parameter integrals agree with original IBP; independent lightlike-bubble 60/100-digit integrals include first invariant derivatives. Their fixtures and reproduction scripts are retained under `validation/pv-high-rank`. Correctly rounded master inputs reduce the synthetic pentagon error to 3.11×10^{-12} with 256-bit reduction and 2.23×10^{-11} with binary64 reduction; this isolates amplified master error. Full independent pentagon evidence follows.

8 Independent pentagon adjudication

These checks establish selected mathematical values independently of an original-reference stability flag. They do not qualify every pentagon tensor, physical graph, runtime domain or numerical precision.

8.1 Rank-six Euclidean tensor

For the retained five-propagator input with squared masses $(4, 9, 16, 25, 36)$, $\mu = 3$ and numerator $q_0^5 q_3$, direct Feynman-simplex integration gives native finite coefficient $-0.0024383789354471 23i$. Two independent simplex coordinate mappings agree within 4.34×10^{-19}; the largest reported quadrature error is 2.77×10^{-15}. The result agrees with the converged 256-bit PV value, and all poles vanish by angular integration and power counting. The positive denominator bound is $F \geq 3.75$. This numerical quadrature uses no tensor-reduction or scalar-master library; its error estimates are not interval certificates. Evidence and the standalone script are in `validation/pv-high-rank/pentagon-simplex-oracle.json`. The original binary64 errors remain 2.23×10^{-10} for Rust and 3.40×10^{-10} for IREGI, above the unchanged gate. The complete public rescue now flags that primary lane unstable and verifies all 210 moments / 840 slots at 128/256 bits, plus an independent absolute-error assertion below 10^{-13} for this coefficient. Evidence is in `validation/rescue-pentagon-rank6.json`. Its retained binary hash and successful assertions establish this run; full output arrays were not printed and the exact source snapshot was not saved at compilation. An isolated Symbolica branch-cache patch also passes all 1320 rank-seven slots at 128/256 bits and this independent coefficient assertion. Its frozen core matches the timed-out baseline; masters and the dependency lock differ. Exact lowering takes 1.784 s in a 115.35 s cold test, not a controlled benchmark. Production integration is pending. The earlier 240 s timeout is retained in `validation/rescue-pentagon-rank7.json`.

8.2 Physical canonical moment 54

For $gg \to HHH$, rank-four moment $q_0 q_3^3$ has UV degree -2; its UV pole must vanish. Original IBP instead returns approximately $3.33691766067 \times 10^{-6} i$ in native normalization. The 128/256-bit PV evaluation suppresses its OneLoop-normalized UV residual from 7.9×10^{-31} to 2.3×10^{-69}, with both IR slots zero. Its converged OneLoop finite coefficient is approximately $3.3474440772594743330 \times 10^{-13} + 2.9254797223941252086 \times 10^{-13} i$. This supports original PV/hybrid and rejects the original IBP result for this component. It is not a full physical tensor comparison. Raw precision evidence is in `validation/pv-high-rank/canonical-pentagon-precision.json canonical-pentagon-adjudication.json` records fixed-gate comparisons.

8.3 Standalone six-dimensional scalar

Use $J_1 = \frac{I_6}{i\pi^3}$, shifts $(0, e_0, e_1, e_2, e_3)$ and squared masses $(10, 12, 15, 18, 22)$. Here $F \geq \frac{39}{4}$, $\det R = 2600$ and $\det G = -16$.

Method	Finite J_1
Independent simplex	$-0.00017714013826400 81806076521310$
Original scalar/general APIs	-0.00014953025024128455; every mode reports stable

Increasing quadrature order and precision through 70 digits converges to the first value. Independent integration of ordinary D4 values over a common mass shift, with a separately integrated tail, agrees within 4.93×10^{-15}. A box-only half-line implementation also agrees to approximately 8.42×10^{-17} or better in the recorded checks. Ordinary D4 and coalesced D4/D6 normalization checks pass. Evidence and reproduction source are in `validation/shifted-pentagon/evidence.json`. The new genuine Rust D6 master passes thirteen consolidated tests: eight genuine-master checks and five endpoint checks, including a common-mass Dual derivative, failed-lane isolation, Float128/ DoubleFloat comparison, numerical guards and independent negative-width agreement at two contour slopes. Public integration now passes a mixed three-lane test: positive-F and negative-width values, rejected endpoint, poles, scale covariance, empty input and internal box count. The unchanged output target is checked after error amplification; the earlier budget failure is retained. See `validation/shifted-pentagon/core-qualification.log`. The public DD/Float batch-3 test timed out at 150 s without a numerical result. Broader endpoints, higher Dual orders and native-feature E6 remain unqualified; no performance claim follows. The separate original complex-box reference discrepancy is documented in `validation/shifted-pentagon/contours.json`.

9 Reference conventions

These conventions were read from the supplied original sources. Their presence in this report is not a claim that every adapter has passed its numerical tests.

9.1 Kinematics and scale

Use Minkowski signature $(+,-,-,-)$ with component order (p^0,p^1,p^2,p^3). Propagators are $D_j = \left(q+p_j\right)^2 - m_j^2 + i0$, raised to positive integer powers. Generalized integrals use $d = 4 + \mathrm{idim} - 2\varepsilon$ and the native raw measure; dimension shifts introduce powers of π. Native tensor reduction accepts one shift per propagator and removes the first common shift internally, then translates tensor coefficients back. The MadLoop interface instead receives the nonzero shifts and inserts a zero first shift. A reference comparison must preserve the chosen numerator routing when it changes the propagator direction.

The original `MU` / `MUR` argument is an energy scale, not its square. Its scalar wrapper passes μ^2 to QCDLoop and μ to OneLOop. Complex mass-squared inputs must retain their imaginary part; the reference switches to its OneLOop backend for complex masses. OneLoopMaster accepts real external squared invariants, complex squared masses with $\Im(m^2) \le 0$, and a positive real squared scale. Its continued logarithms and polylogarithms own the Feynman branch prescription; no independent branch implementation belongs in the reduction adapter. Preserve inputs by default; legacy on-shell snapping must be requested explicitly. Dual derivatives follow the selected analytic branch away from nondifferentiable boundaries.

9.2 Native poles and MadLoop conversion

The native result order is

$$(U,I,J,F) = (\varepsilon_{\mathrm{UV}}^{-1}, \varepsilon_{\mathrm{IR}}^{-1}, \varepsilon_{\mathrm{IR}}^{-2}, \mathrm{finite}).$$

These symbols label separate coefficients. In particular, a scaleless integral can have canceling UV and IR contributions even when its combined Laurent series is zero. Three combined coefficients do not determine their separate origin. OneLoopMaster uses $(\mathrm{finite}, \varepsilon^{-1}, \varepsilon^{-2})$ ordering.

The original `IMLOOP` adapter combines the native poles and maps a tensor of homogeneous rank r to MadLoop order (finite, simple pole, double pole). Defining $a = \gamma_E + \ln\pi$, $b = \frac{a^2}{2} + \frac{\pi^2}{12}$ and $c_r = \frac{(-1)^r}{i\pi^2}$, its explicit mapping is

$$(F_{\mathrm{ML}}, S_{\mathrm{ML}}, D_{\mathrm{ML}}) = c_r(F + a(U+I) + bJ, U + I + aJ, J).$$

For a OneLOop-normalized scalar master (L_0, L_1, L_2), the scalar wrappers use

$$J = i\pi^2 L_2, \quad U + I = i\pi^2(L_1 - aL_2),$$

$$F = i\pi^2\left(L_0 - aL_1 + \left(\frac{a^2}{2} - \frac{\pi^2}{12}\right)L_2\right).$$

The UV/IR split requires independent integral classification. In particular, the scaleless bubble is $(U,I,J,F) = (i\pi^2, -i\pi^2, 0, 0)$ in the original wrapper. Applying the MadLoop conversion to an already normalized master result would count normalization twice.

9.3 Polynomial and tensor interfaces

A rank bound R has $\binom{R+4}{4}$ Cartesian coefficients in the original four-component basis. `ti_reduce::calc_pos` traverses total degree in ascending order, then exponents of q^0, q^1, q^2 in descending order; the exponent of q^3 is the remaining degree. Thus the start is $\left(1, q^0, q^1, q^2, q^3, (q^0)^2, q^0q^1, q^0q^2, q^0q^3, (q^1)^2, \ldots\right)$. Numerator preparation and tensor output must share this order. The original native and MadLoop tensor conventions differ in multinomial factors: the native path multiplies symmetric components by the factor returned by `all_integers`, whereas `ML5_CONVENTION=.TRUE.` uses unweighted components. The independent oracle sets this flag **without** applying `IMLOOP`, preserving the native four-pole normalization. Mixed-monomial tests detect multiplicity mismatches.

Reduction modes are 0: IBP, 1: PaVe, 2: PaVe with IBP stability improvement. Direction mode 2 reverses the propagator order and shifts the loop origin. The source labels direction modes 4/5 as unimplemented quadruple precision; their numbers are not evidence of high-precision support.

10 Algebra reuse audit

All symbolic algebra belongs to Symbolica and the FeynKit / Spenso / Idenso ecosystem. Before introducing a missing operation, record three checks: Symbolica APIs and implementation; Symbolica usage/examples/tests; and FeynKit/gammaloop facilities and tests. An adapter or data-layout conversion must not grow an independent polynomial, simplification or matrix engine.

10.1 Existing facilities to reuse

- **Tensor identities and contractions:** Idenso contracts metrics and reduces supported gamma/color identities; Spenso carries tensor structure. Keep Lorentz dimension and any gamma-five prescription explicit.
- **Coefficient extraction:** FeynKit's tensor reducer uses Idenso contraction without global expansion, `to_dots()` and the shared `coefficient_list` collector. Preserve scalar factorization until polynomial extraction requires expansion.
- **Propagator and scalar-product algebra:** FeynKit's `IntegralFamily` uses Symbolica for polynomial conversion, exact rank and linear solves, basis completion and mappings. Its Gaussian-rational conversion avoids statistical expression-field zero tests that may discard small coefficients.
- **Numerical preparation:** Compile reusable Symbolica expressions outside the sample loop. Evaluation and scalar-master types must preserve the selected precision and all Dual components without binary64 round trips.
- **Laurent coefficients:** IBP/PV factors and native normalization use cached Symbolica `series` templates and `coefficient` extraction. The three reuse checks are the pinned Symbolica `src/atom/core.rs` and `src/poly/series.rs` APIs; its `src/derivative.rs` pole-series tests; and existing use in `vakint/src/projection.rs` and OneLoopMaster's `src/sheet_exact.rs`. No separate Laurent convolution engine is needed. Symbolica also evaluates the independent Gamma/series triangle oracle.

10.2 HEPKit tensor-reduction optimizations

The local `feynkit-tensor` reducer stores repeated-vector contractions as symmetry orbits instead of factorially many labeled pairings. Its joint-color dynamic program builds orbit intersections; alternating cycles build orbit Gram matrices; Symbolica solves the smaller orbit system. A rank-20 isotropic projection can have one orbit instead of 19!! labeled pairings.

The implementation caches canonical Weingarten coefficients and has a closed isotropic path. General projectors are validated through rank 20; the special all-equal isotropic route reaches rank 32. These are upstream limits, not a statement of iregi-rust's current supported domain.

For denominators with external momenta, declare an independent external basis and average only the transverse space. Odd total tensor ranks need not vanish. Keep D symbolic while constructing high-rank projectors; early integer specialization can make the universal metric basis singular. External Gram inverses can amplify small-Gram numerical errors, so projector preprocessing requires a conditioning-aware comparison against direct recurrences.

This projector reduces tensor numerators, not complete scalar integral families. Reuse it where applicable and benchmark the resulting plan; no factorial-count argument alone establishes a numerical speedup. Retain explicit output/pairing limits and report their failures honestly.

10.3 References and provenance

The untouched generalized IREGI entry point stops during factorial-table initialization: the configured powers/rank request factorial 16, while the helper rejects arguments above 12. A narrow repair in a generated reference copy replaces that table expression by its exact binomial recurrence. The untouched/repaired twins agree exactly on 24 ordinary cases; all 80 table entries match independent integer binomials; seven generalized fixtures check normalization, unit powers, mass derivatives and a tadpole dimension shift. The original `STOP` returns exit code zero, so the harness also requires an explicit result record. Evidence lives in `validation/reference/fixtures.json` and the report's JSON snapshot.

OneLoopMaster documents corrected analytic-continuation defects in its upstream OneLOop reference, including complex-mass box cases. Preserve each disagreement as a visible case and adjudicate it with independent analytic, high-precision or parameter-integral evidence. Agreement must not mean reproducing a known incorrect branch.

Primary references: Alwall et al., appendix C.1 HEPKit TensorReducer Goode et al., tensor projectors Collins–Matsumoto, Weingarten recurrence NinjaRS report style. Local source audit: FeynKit's `feynkit-tensor/src` projector modules and `feynkit-graph/src/integrals.rs` original IREGI's `IREGI4ML5_interface.f90`, `ti_reduce.f90`, `si_reduce.f90`, `mis_warp.f90` and `funlib.f90`.